CreditRisk+

CreditRisk+ is a credit risk management model developed by Credit Suisse, primarily focused on estimating potential losses in a credit portfolio due to defaults. Unlike traditional credit risk models, which may rely heavily on historical default data, CreditRisk+ is based on an actuarial approach, using statistical methods commonly seen in insurance.

1. Model Foundation and Key Concepts

- **Actuarial Approach**: CreditRisk+ uses **Poisson processes** to model default events, assuming that the number of defaults follows a **Poisson distribution**. This allows for flexible modelling even when historical data on defaults is limited.
- **Default Probability**: It treats defaults as statistically independent events, focusing on default probabilities rather than specific causes or timing of defaults.
- **Loss Distribution**: CreditRisk+ estimates a **loss distribution** for a credit portfolio by aggregating the probability of default across all loans or exposures within the portfolio.

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2. How CreditRisk+ Works

- **Step 1: Calculating Default Frequencies**
  Each credit exposure (or borrower) has an assigned probability of default, often based on credit rating. This probability determines the **frequency** of default for each borrower.
- **Step 2: Calculating Loss Severity**
  CreditRisk+ assumes that the severity of loss (or exposure amount in case of default) is fixed for each borrower, simplifying the loss calculation. These loss amounts are used in the aggregation of the loss distribution.
- **Step 3: Aggregating Loss Distribution**
  Using the Poisson model, CreditRisk+ calculates an aggregate **loss distribution** by combining the default frequencies and loss severities across the portfolio. This distribution represents possible outcomes for the total portfolio loss due to defaults.
- **Step 4: Stress Testing and Sector Analysis**
  CreditRisk+ allows for **sector-based analysis**, where portfolios can be divided into different sectors (e.g., industries, regions), with each sector assigned a different likelihood of default. This enables stress testing to see how a portfolio would respond to adverse conditions in specific sectors.

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3. Assumptions of CreditRisk+

- **Independence of Defaults**: Defaults are assumed to be independent events, meaning the model does not account for contagion effects where one default might trigger others.
- **Fixed Loss Given Default**: The model assumes a fixed exposure in case of default for each borrower.
- **Poisson Default Modeling**: It assumes defaults follow a Poisson distribution, which works well for modelling rare events with a given average rate of occurrence.

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4. Applications of CreditRisk+

- **Portfolio Risk Management**: Used by banks and financial institutions to assess and manage the risk of default across their credit portfolios.
- **Capital Allocation**: Helps in determining the capital reserves required to cover potential credit losses.
- **Scenario and Sensitivity Analysis**: CreditRisk+ is useful for testing the impact of different economic scenarios or stress conditions, as it allows for sector-specific risk assessments.

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5. Limitations of CreditRisk+

- **No Correlation Between Defaults**: By assuming independence between defaults, CreditRisk+ may underestimate risk in times of economic stress or crises when defaults can be correlated.
- **Fixed Loss Severity**: It does not account for varying losses given default (LGD), which may reduce accuracy in portfolios with significant LGD variation.
- **Simplified Assumptions**: The model's simplicity means it may miss nuanced risk factors like credit quality changes over time or exposure growth.

Summary of CreditRisk+

CreditRisk+ is a practical, actuarial-based model for estimating credit risk, particularly for portfolios where historical data is limited. It's best suited for high-level portfolio risk estimation and capital allocation rather than predicting exact defaults or losses in highly correlated, volatile portfolios.

Basic Python Code for Expected Loss and Poisson Loss Distribution

Below is a Python example to compute the expected loss for a portfolio and to generate a Poisson-based loss distribution.

```
import numpy as np
from scipy.stats import poisson

# Define data for each borrower in the portfolio
borrowers = [
    {"EAD": 100000, "PD": 0.01},  # Borrower 1
    {"EAD": 50000, "PD": 0.02},   # Borrower 2
    {"EAD": 200000, "PD": 0.015}  # Borrower 3
]

# Calculate Expected Loss for each borrower
expected_losses = [borrower["EAD"] * borrower["PD"] for borrower in borrowers]
total_expected_loss = sum(expected_losses)
print(f"Total Expected Loss for the Portfolio: {total_expected_loss:.2f}")

# Set up loss distribution using Poisson approximation for simplicity
# Sum PDs to get average portfolio default frequency (lambda)
lambda_portfolio = sum(borrower["PD"] for borrower in borrowers)

# Generate Poisson distribution of defaults for the portfolio
# Number of defaults we want to consider in the distribution
num_defaults = np.arange(0, 11)  # For 0 to 10 defaults

# Calculate Poisson probabilities for each number of defaults
poisson_probs = poisson.pmf(num_defaults, lambda_portfolio)
print("\nPoisson Probabilities for Number of Defaults:")
for k, prob in zip(num_defaults, poisson_probs):
    print(f"Pr({k} defaults) = {prob:.4f}")

# Calculate potential loss amounts for each default scenario
loss_per_default = sum(borrower["EAD"] for borrower in borrowers) / len(borrowers)  # average
EAD
```

```
potential_losses = loss_per_default * num_defaults  # total potential loss at each default
level

print("\nPotential Loss Amounts:")
for k, loss in zip(num_defaults, potential_losses):
    print(f"Loss if {k} defaults: {loss:.2f}")
```

1. Core Formulas in CreditRisk+

A. Expected Loss (EL)

The expected loss for a single borrower is calculated as:

$$\mathrm{EL}_i = \mathrm{EAD}_i \times \mathrm{PD}_i$$

where:

- EAD_i = Exposure at Default for borrower i
- PD_i = Probability of Default for borrower i

B. Loss Distribution Using Poisson Distribution

1. **Individual Default Loss Distribution**:
   For a single borrower, the probability of k defaults in a time period is given by the Poisson probability:

   $$P(X_i = k) = \frac{(\lambda_i)^k \cdot e^{-\lambda_i}}{k!}$$

   where:

   - $\lambda_i = \mathrm{PD}_i$, the average number of defaults in a period for borrower i
   - X_i represents the random variable for the number of defaults
2. **Portfolio Loss Distribution**: The portfolio's total loss distribution aggregates the default probabilities across all borrowers. The **convolution** method is commonly used to sum up these individual Poisson distributions, which requires specialized techniques in practice.
3. **Sector-Based Loss Modeling**:
   If the portfolio is segmented into sectors, we assign each sector a risk factor and calculate the sector's loss distribution accordingly. Then, total portfolio risk is the weighted sum of sector risks.